volatility_filter

equity strategy | family: equity-volatility | status: research | derived from `balanced_leverage`

What this strategy does

Balanced Leverage, skipping long/short x jumpiness groups that have not made money so far

Why it should work. Skip the cells that history says do not pay -- but the data shows a gradient, not a bad cell.

Design notes (from the strategy's source file).

Volatility Filter (and Calm Weighted Filter) -- skip stock groups that have not paid, judged only on past data.

Drops a (side, idio-vol tercile) cell from new entries when its own history says it has not earned its keep, judged ONLY on quarters whose positions had fully exited. Result: filtering does not help on its own (corrected run: Sharpe 0.89 vs 1.09 for Balanced Leverage; pre-fix 1.23 vs 1.30; the effect is a smooth gradient, not one broken cell) and adds nothing once inverse-vol sizing is on (corrected: Calm Weighted Filter 1.08 < Calm Weighted 1.24; pre-fix 1.41 < 1.47). Kept as a documented negative result: size by vol, do not filter by it.

parameter	value
base_unit_fraction	0.003
max_gross_leverage	2.5
liquidity_cap_frac	0.05
return_basis	mktadj
cap_mode	priority
exposure_cap	open_book
beta_overlay	0.0

Performance

metric	volatility filter	market (CRSP VW)
Annualized return	4.60%	9.94%
Annualized volatility	5.20%	19.35%
Sharpe (NAV-pct)	0.89	0.59
Sortino	1.27	
Max drawdown	-18.8%	-55.5%
Calmar	0.25	
Final NAV ($10M start)	$24,019,423	
Worst day	-4.29%	
% positive days	49.9%	
Sharpe 1996-2001	1.43	
Sharpe 2002-2007	1.07	
Sharpe 2008-2013	0.19	

Correlation of daily returns with the market: **0.03**.

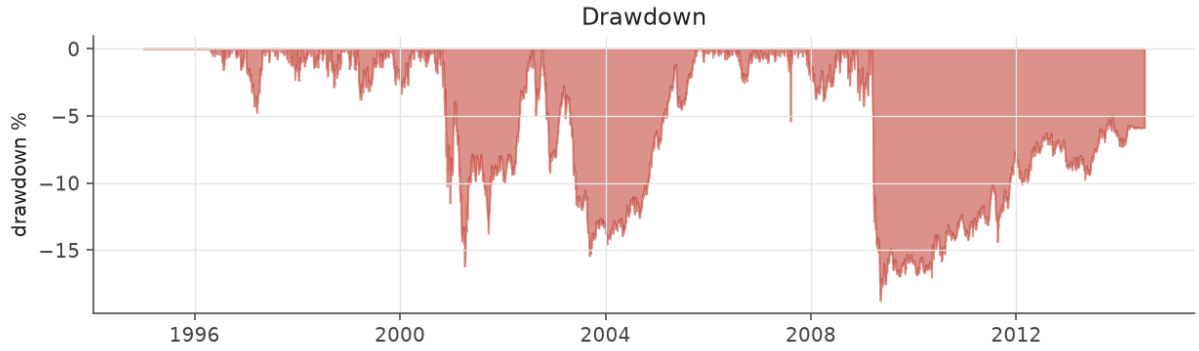
• best Sharpe among equity: `calm_rank_tilt` (1.47); this one: 0.891

• best return among equity: `market_boost_max` (0.116); this one: 0.046

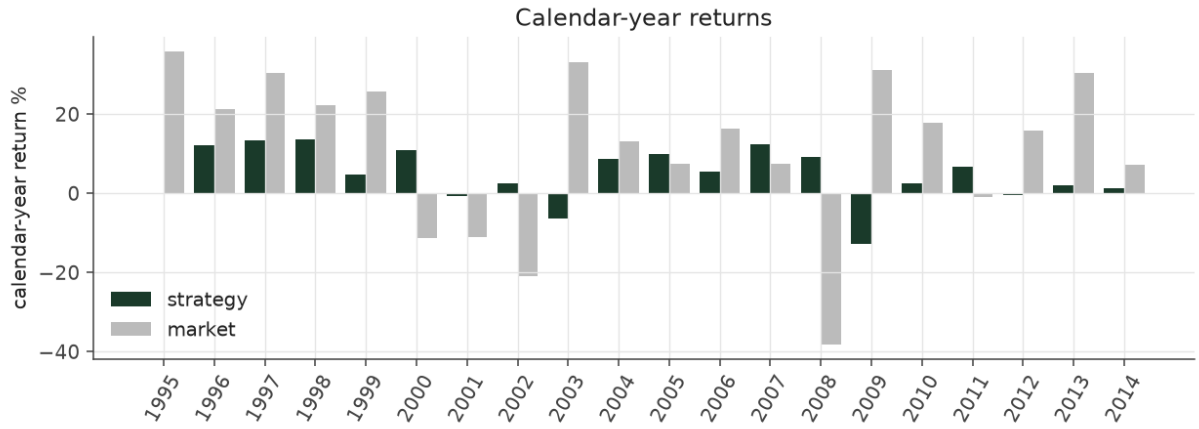
• best Calmar among equity: `calm_rank_tilt` (0.442); this one: 0.245



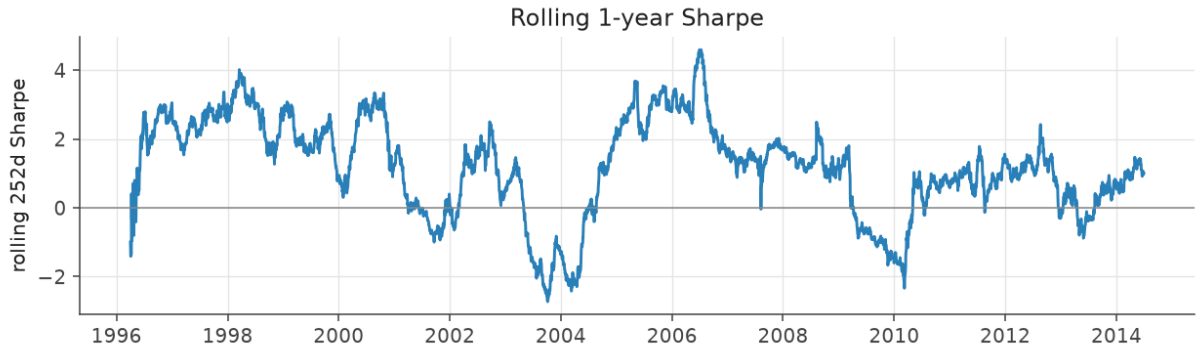
NAV (log scale)



Drawdown from running peak

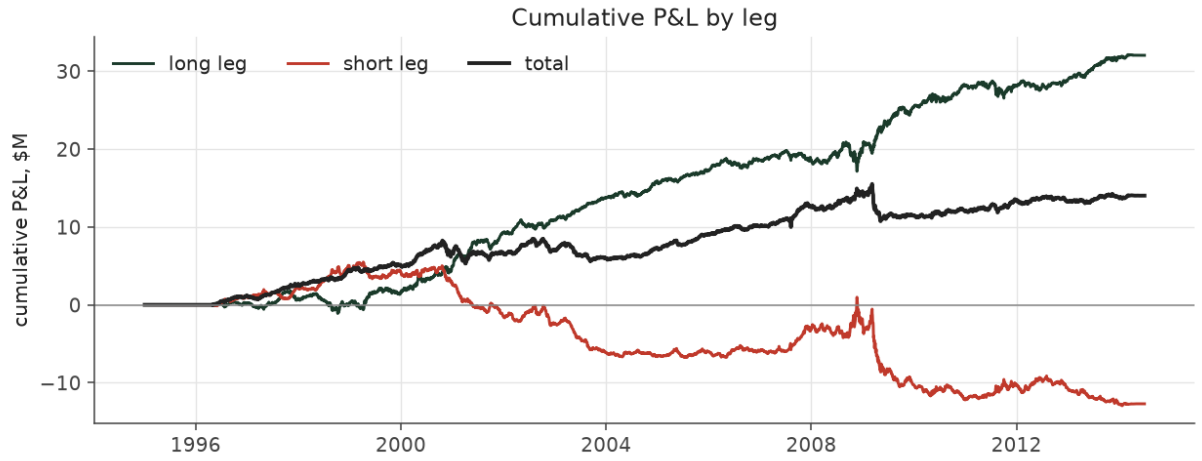


Calendar-year returns



Is the edge stable, or concentrated in a few years?

Where the return comes from



Long leg vs. short leg vs. total

	long leg	short leg
P&L ($M, before costs)	32.04	-12.71
P&L after costs ($M)	29.45	-15.43
of which common (average-event) component ($M)	20.72	-21.05
of which excess / PEAD component ($M)	11.31	8.34
excess minus costs ($M)	8.72	5.62
gross return per $ traded (bp)	247.91	-96.95
positions in the money (hit rate)	0.56	0.48
stand-alone Sharpe (leg only)	1.17	-0.38

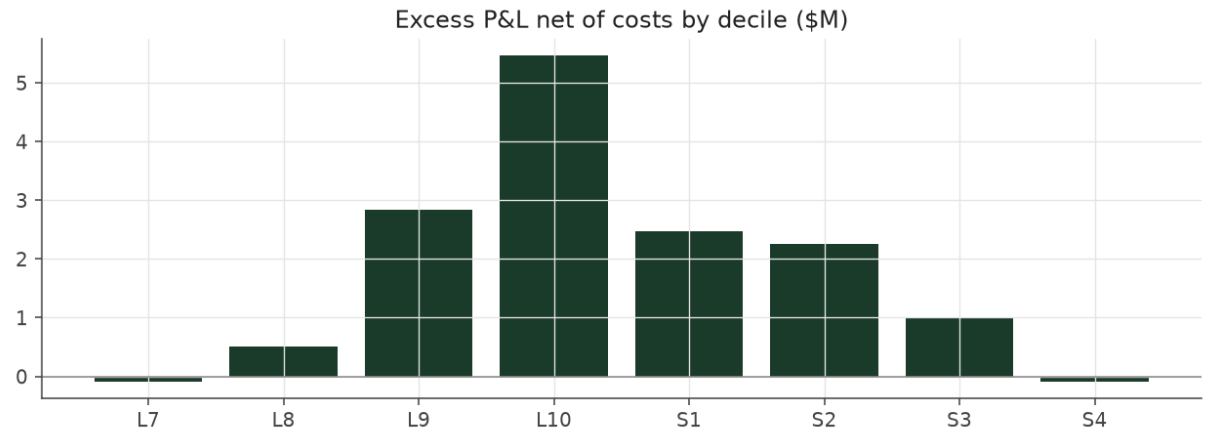
Daily P&L correlation between the two legs: **-0.78**.

• **The short leg loses money in absolute terms** (-12.7M before costs) while the long leg earns +32.0M. That is mostly a benchmark effect, not a signal failure: the average earnings-event stock beat the cap-weighted market during its holding window, which flatters the long leg and hurts the short leg by 20.7M / 21.1M. Stripping that common component, the surprise signal itself earned +11.3M on the long side and +8.3M on the short side.

• The two legs' daily P&L are -0.78 correlated: the short leg is the hedge that makes the long leg's variance disappear (stand-alone leg Sharpe 1.17 long / -0.38 short vs 0.89 combined).

By decile and side (signal-driven part only)

leg	decile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
long	7	5,837	84.34	1.35	0.08	0.17	-0.10	-0.7%	-11.29
long	8	15,020	282.62	5.69	1.08	0.57	0.51	3.6%	18.14
long	9	17,362	397.36	8.31	3.63	0.79	2.84	19.8%	71.48
long	10	19,198	527.91	14.10	6.53	1.06	5.46	38.1%	103.53
short	1	17,107	522.20	-6.32	3.59	1.12	2.47	17.2%	47.22
short	2	17,394	451.29	-5.08	3.17	0.92	2.25	15.7%	49.92
short	3	13,787	300.06	-3.45	1.60	0.60	0.99	6.9%	33.15
short	4	2,168	37.34	-0.58	-0.02	0.07	-0.09	-0.6%	-23.82



D10 and the short deciles carry the alpha; the middle of each leg earns little

By size quintile (1 = smallest)

size quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1	12,911	253.77	0.22	1.74	0.98	0.76	5.3%	30.11
2	20,633	496.97	5.33	6.97	1.40	5.57	38.8%	112.09
3	23,152	592.28	3.80	5.00	1.25	3.75	26.1%	63.31
4	25,014	626.51	2.96	3.73	0.98	2.74	19.1%	43.78
5	26,163	633.59	1.70	2.22	0.70	1.52	10.6%	23.99

By liquidity quintile (1 = least liquid)

liquidity quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1	12,343	188.03	2.26	3.54	0.94	2.60	18.1%	138.26
2	21,074	533.40	3.17	5.47	1.60	3.87	27.0%	72.63
3	23,523	634.05	4.44	5.86	1.27	4.60	32.0%	72.49
4	24,959	628.11	2.99	3.40	0.88	2.52	17.6%	40.12

5	25,970	619.37	1.15	1.36	0.62	0.74	5.2%	11.99

By holding-window length

hold bucket	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
<=40d	38,259	889.88	-0.30	6.33	1.71	4.63	32.3%	52.00
41-60d	25,572	650.73	1.94	4.69	1.44	3.25	22.7%	50.02
61-90d	35,161	855.70	7.46	4.41	1.58	2.83	19.7%	33.01
>90d	8,881	206.81	4.93	4.22	0.58	3.64	25.4%	175.99

By era

era	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1996-01	31,383	620.57	6.69	7.87	1.23	6.64	46.3%	107.05
2002-07	34,703	870.62	5.61	7.32	1.74	5.58	38.9%	64.09
2008-13	41,787	1,111.94	1.71	4.47	2.34	2.12	14.8%	19.11

By FF12 sector

ff12 sector	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
BusEq	18,723	246.30	0.19	0.65	0.52	0.12	0.9%	5.08
Chems	2,528	177.05	0.67	1.01	0.32	0.69	4.8%	39.15
Durbl	2,772	192.97	1.35	1.80	0.36	1.43	10.0%	74.24
Enrgy	6,145	218.16	2.46	2.89	0.42	2.47	17.2%	113.13
Hlth	10,525	246.44	0.98	1.50	0.52	0.97	6.8%	39.58
Manuf	11,344	232.51	1.16	1.72	0.49	1.23	8.5%	52.70
Money	15,356	215.07	1.88	2.45	0.50	1.95	13.6%	90.62
NoDur	4,848	214.58	0.86	1.23	0.42	0.82	5.7%	38.03
Other	16,089	235.74	1.53	2.10	0.53	1.57	11.0%	66.72
Shops	11,147	243.84	2.49	3.00	0.51	2.49	17.3%	102.04
Telcm	3,731	212.11	0.70	1.07	0.40	0.67	4.7%	31.62
Utils	4,665	168.35	-0.24	0.25	0.32	-0.07	-0.5%	-4.10

By idiosyncratic-volatility tercile

vol tercile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
high vol	33,987	765.25	-0.67	1.76	1.41	0.35	2.5%	4.60
low vol	36,925	910.29	7.29	9.27	2.03	7.24	50.4%	79.50
mid vol	35,223	879.88	6.62	7.78	1.76	6.02	42.0%	68.45

Short-borrow sensitivity (not modeled in the backtest)

borrow rate	fee M	ann. return	drag
0.0%	0.0	4.60%	-0.00%
0.5%	1.3	4.30%	0.30%
1.0%	2.6	3.98%	0.62%
2.0%	5.3	3.28%	1.33%
5.0%	13.2	0.40%	4.21%

The engine charges the same transaction cost on shorts as on longs and no stock-borrow fee. The short leg is concentrated in small, illiquid names, where borrow is scarce and expensive -- read the table as a floor on the cost of running the short leg.